

10(a)	(melanic x melanic) Aa x Aa (Gametes) A a and A a ; (Offspring) AA Aa Aa aa ; (melanic, melanic, melanic, typical) (Ratio) 3 melanic : 1 typical ;	3									
10(b)	it / (melanic) allele, is dominant ;	1									
10(c)(i)	melanic / dominant phenotype, is crossed with homozygous recessive ;	1									
10(c)(ii)	<table><tr><td></td><td>same locus (A / a)</td><td>different loci (A / a and B / b)</td></tr><tr><td>genotypes of melanic moths from cross 1</td><td>AA Aa (aA) ;</td><td>AaBb Aabb aaBb ;</td></tr><tr><td>proportion of test crosses (cross 2) giving 100% melanic offspring</td><td>1 in 3</td><td>none / 0 ;</td></tr></table>		same locus (A / a)	different loci (A / a and B / b)	genotypes of melanic moths from cross 1	AA Aa (aA) ;	AaBb Aabb aaBb ;	proportion of test crosses (cross 2) giving 100% melanic offspring	1 in 3	none / 0 ;	3
	same locus (A / a)	different loci (A / a and B / b)									
genotypes of melanic moths from cross 1	AA Aa (aA) ;	AaBb Aabb aaBb ;									
proportion of test crosses (cross 2) giving 100% melanic offspring	1 in 3	none / 0 ;									
10(d)	(24 × 29) ÷ 8 ; = 87 ;	2									